

Evaluation of an explicit local integral

Complete formalized statement

Version 1

This document specifies all data, definitions and hypotheses needed for the formalized theorem. The function Φ_0 is defined by an explicit finite formula below.

1 Field, measures and scalar parameters

Let F be a nonarchimedean local field. Equip it with its local-field topology and Borel σ -algebra, and equip $F^\times$ with the induced topology and Borel structure. Let

$$v : F \longrightarrow \mathbb{Z} \cup \{+\infty\}$$

be its normalized additive valuation, defining the local-field topology: $v(0) = +\infty$ and $v(F^\times) = \mathbb{Z}$. Put

$$\mathcal{O} = \{x \in F : v(x) \geq 0\}, \quad \mathfrak{m} = \{x \in F : v(x) > 0\}, \quad q = \#(\mathcal{O}/\mathfrak{m}).$$

Assume $q > 2$. Fix $\varpi \in F^\times$ with $v(\varpi) = 1$, and set

$$|a|_F = q^{-v(a)} \quad (a \in F^\times).$$

Let $\sqrt{q}$ denote the positive real square root. Let μ be the normalized additive Haar measure on F with $\mu(\mathcal{O}) = 1$, and let ν be the normalized multiplicative Haar measure on $F^\times$ with

$$\nu(\mathcal{O}^\times) = (1 - q^{-1}).$$

All integrals below use the product measure $\lambda = \nu \otimes (\mu \otimes \mu)$ on $F^\times \times (F \times F)$.

Fix complex parameters α, β satisfying

$$|\alpha| = |\beta| = 1, \quad \alpha \neq 1, \quad \beta \neq 1, \quad \alpha \neq \beta, \quad \alpha\beta \neq 1. \tag{1}$$

2 The eight coefficients and the explicit kernel

Define

$$C_q = 1 + 2q^{-1} + 2q^{-2} + 2q^{-3} + q^{-4}, \quad R(z) = \frac{1 - q^{-1}z}{1 - z}. \quad (2)$$

For each $0 \leq i \leq 7$, define complex numbers $z_{i1}, \dots, z_{i4}$, U_i, V_i by the following table, and put $A_i = \prod_{j=1}^4 R(z_{ij})$.

i	$(z_{i1}, z_{i2}, z_{i3}, z_{i4})$	U_i	V_i
0	$(\beta\alpha^{-1}, \beta^{-1}, \alpha^{-1}\beta^{-1}, \alpha^{-1})$	$\alpha\beta$	α
1	$(\alpha\beta^{-1}, \alpha^{-1}, \alpha^{-1}\beta^{-1}, \beta^{-1})$	$\alpha\beta$	β
2	$(\alpha^{-1}\beta^{-1}, \beta, \beta\alpha^{-1}, \alpha^{-1})$	$\alpha\beta^{-1}$	α
3	$(\alpha\beta, \alpha^{-1}, \beta\alpha^{-1}, \beta)$	$\alpha\beta^{-1}$	β^{-1}
4	$(\alpha^{-1}\beta^{-1}, \alpha, \alpha\beta^{-1}, \beta^{-1})$	$\beta\alpha^{-1}$	β
5	$(\alpha\beta, \beta^{-1}, \alpha\beta^{-1}, \alpha)$	$\beta\alpha^{-1}$	α^{-1}
6	$(\beta\alpha^{-1}, \alpha, \alpha\beta, \beta)$	$\alpha^{-1}\beta^{-1}$	β^{-1}
7	$(\alpha\beta^{-1}, \beta, \alpha\beta, \alpha)$	$\alpha^{-1}\beta^{-1}$	α^{-1}

The hypotheses (1) make every denominator in this table's definition of A_i nonzero.

For $a \in F^\times$ and $x, y, z \in F$, form the matrix

$$g = g(a; x, y, z) = \begin{pmatrix} 1 & 0 & y & z \\ 0 & a & ax & ay \\ 0 & 0 & a^{-1} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (3)$$

For such a matrix g , define

$$u(g) = \min_{1 \leq i, j \leq 4} v(g_{ij}), \quad (4)$$

$$s(g) = \min_{\substack{1 \leq i_1 < i_2 \leq 4 \\ 1 \leq j_1 < j_2 \leq 4}} v(g_{i_1 j_1} g_{i_2 j_2} - g_{i_1 j_2} g_{i_2 j_1}). \quad (5)$$

The minimum in (5) is over all 36 two-by-two minors. The valuation of a zero entry or minor is $+\infty$. Both displayed minima are finite integers. Put

$$n(g) = u(g) - s(g), \quad B(g) = s(g) - 2u(g), \quad \ell(g) = 2n(g), \quad H(g) = n(g) + B(g) = -u(g). \quad (6)$$

Finally, *define* the complex-valued function on these matrices by

$$\Phi_0(g(a; x, y, z)) = \frac{1}{C_q} \sum_{i=0}^7 A_i (q^{-1/2})^{6n(g)+4B(g)} U_i^{n(g)} V_i^{B(g)}. \quad (7)$$

The notation Φ_0 in this document means precisely (7).

3 Integrands and the closed expression

Set

$$(z_0, z_1, z_2) = (0, \varpi^{-1}, \varpi^{-2}), \quad (c_0, c_1, c_2) = ((1 - q^{-1})^{-1}, q, -q/(1 - q^{-1})). \quad (8)$$

For $\delta \in \mathbb{C}^\times$, a real number $T > 0$, and $r \in \{0, 1, 2\}$, define the function $f_{r,T,\delta}$ on $F^\times \times F \times F$ by

$$f_{r,T,\delta}(a, x, y) = \delta^{v(a)} (q^{-v(a)})^{3/2} T^{H(g(a;x,y,z_r))} \Phi_0(g(a; x, y, z_r)). \quad (9)$$

Whenever the function in (9) is *absolutely integrable*, put

$$\mathcal{I}_T(\delta) = \sum_{r=0}^2 c_r \int_{F^\times \times (F \times F)} f_{r,T,\delta} d\lambda. \quad (10)$$

The theorem below proves absolute integrability on every parameter range where it uses (10).

Let $\mathcal{A} = (\alpha, \alpha^{-1}, \beta, \beta^{-1})$, regarded as an ordered tuple so that products retain multiplicities. Define

$$N_1 = \prod_{\zeta \in \mathcal{A}} (1 - \zeta q^{-1}), \quad (11)$$

$$N_2 = (1 - \alpha\beta q^{-1})(1 - \alpha^{-1}\beta q^{-1})(1 - \alpha\beta^{-1} q^{-1})(1 - \alpha^{-1}\beta^{-1} q^{-1}), \quad (12)$$

$$D_1(\delta) = \prod_{\zeta \in \mathcal{A}} (1 - \zeta \delta (q^{-1/2})), \quad (13)$$

$$D_2(\delta) = \prod_{\zeta \in \mathcal{A}} (1 - \zeta \delta^{-1} (q^{-1/2})). \quad (14)$$

Whenever its denominator is nonzero, define the quotient

$$E_q(\alpha, \beta, \delta) = \frac{\frac{(1 + \alpha)^2 (1 + \beta)^2}{\alpha \beta} (1 - q^{-1}) N_1 N_2}{C_q D_1(\delta) D_2(\delta)}. \quad (15)$$

All denominators used in the theorem are proved nonzero under its hypotheses.

Every limit denoted by $T \uparrow 1$ below is a limit through *real* T satisfying $0 < T < 1$.

4 The formalized theorem

Theorem 1. *For every $\delta \in \mathbb{C}$ and real T with*

$$q^{-1/2} \leq |\delta| \leq 1, \quad 0 < T \leq 1, \quad T < |\delta|q^{1/2}, \quad (16)$$

all three functions $f_{r,T,\delta}$, for $r \in \{0, 1, 2\}$, are measurable and absolutely integrable with respect to λ , and so $\mathcal{I}_T(\delta)$ is well defined and finite. In particular, this holds for every $0 < T < 1$ on $q^{-1/2} \leq |\delta| \leq 1$, and for $T = 1$ on $q^{-1/2} < |\delta| \leq 1$.

Moreover, the following identities hold.

1. *If $(q^{-1/2}) < |\delta| \leq 1$, then $C_q D_1(\delta) D_2(\delta) \neq 0$ and*

$$\boxed{\lim_{T \uparrow 1} \mathcal{I}_T(\delta) = \mathcal{I}_1(\delta) = E_q(\alpha, \beta, \delta).} \quad (17)$$

2. *Let $\varepsilon \in \{1, -1\}$ and assume additionally*

$$\alpha \neq \varepsilon, \quad \beta \neq \varepsilon. \quad (18)$$

Then $C_q D_1(\varepsilon(q^{-1/2})) D_2(\varepsilon(q^{-1/2})) \neq 0$, and we have

$$\boxed{\lim_{T \uparrow 1} \frac{1 - \varepsilon}{q + 1} \mathcal{I}_T(\varepsilon(q^{-1/2})) = \frac{1 - \varepsilon}{q + 1} E_q(\alpha, \beta, \varepsilon(q^{-1/2})).} \quad (19)$$

Remark 2. *In part (iv) of Theorem 1, the case $\varepsilon = 1$ is trivial and simply states $0 = 0$. In the case $\varepsilon = -1$, formula (19) is the assertion*

$$\lim_{T \uparrow 1} \frac{2}{q + 1} \mathcal{I}_T(-(q^{-1/2})) = \frac{2}{q + 1} E_q(\alpha, \beta, -(q^{-1/2})).$$

The theorem does not assert the existence of $\mathcal{I}_1(\delta)$ at either endpoint $\delta = (q^{-1/2})$ or $\delta = -(q^{-1/2})$.

Remark 3. *There are no additional analytic or representation-theoretic hypotheses behind Theorem 1. All definitions and assumptions used are stated in this document.*